Comparing Performance Patterns in Tested vs. Untested Powerlifting Competition Entries

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1. Introduction and Motivation

Few questions in strength sports generate more debate than whether an athlete competes with or without performance-enhancing drugs (PEDs), a topic commonly framed as "natty or not?" in gym culture. Competitive powerlifting provides a natural setting for studying this question with data, because federations and meets vary in whether they require drug testing. The OpenPowerlifting public dataset (OpenPowerlifting, 2026), containing over 3.8 million competition records, captures this distinction through a "Tested" field that marks whether a given entry counted as drug-tested. Understanding how performance differs across these contexts could help athletes and coaches set realistic benchmarks for specific competition situations.

2. Dataset Description

2.1 Source

The OpenPowerlifting bulk CSV file is a public, community-maintained archive of powerlifting competition results. The raw file contains 3,868,570 rows and 42 columns, where each row is one competition entry (one lifter's result from one meet). This dataset is appropriate because it records the tested/untested distinction at the entry level, the variable the research questions are built around.

2.2 Key Features

Performance is measured by `TotalKg` (sum of best squat, bench, and deadlift) and `Dots` (a bodyweight-normalized score; Kopayev et al., 2020). Entries are grouped by `Sex`, `Age`, `Equipment` (Raw, Wraps, Single-ply, Multi-ply), and `WeightClassKg`. The `Tested` field marks whether the entry counted as drug-tested, not whether the lifter was actually tested. Four additional fields (`Event`, `Division`, `Place`, `AgeClass`) were used only during cleaning.

2.3 Preprocessing

Only Open-division entries with all three lifts (squat, bench, deadlift), from male or female competitors in a standard equipment type (Raw, Wraps, Single-ply, Multi-ply) with a valid total were kept. Disqualifications, drug test disqualifications, no-shows, and anyone under 18 were removed. The `Tested` field was converted to a true/false flag, and lifters whose names appeared on both sides of the tested/untested divide (crossover lifters) were dropped so the two groups would not share members.

Early exploratory data analysis (EDA) that shaped the research questions kept 570,847 rows, accepted more kinds of Open divisions, did not drop crossover lifters, and only included Raw and Wraps entries. The data cleaning steps outlined above produced a cleaned base dataset with 510,882 rows, split roughly two-thirds tested and one-third untested.

2.4 Challenges

`Age` is missing in 37.1% of raw rows and 28.7% of cleaned rows, which reduces the sample available for RQ3. The oldest age bins (especially 61+) have the fewest rows, making their bootstrap ranges wider and their evidence less precise.

3. Research Questions / Objectives

RQ1: Main Performance Comparison

Do untested entries tend to show higher TotalKg and DOTS scores than tested entries? Comparisons are grouped by sex, weight class, and equipment so that each matchup is apples-to-apples.

RQ2: Tested-Entry Share by Performance Level

Is tested status associated with DOTS-score bin, separately by sex? Entries are divided into equal-width DOTS bins, and the tested-entry share is compared across bins.

RQ3: Age Pattern Comparison

Do untested entries tend to show higher mean TotalKg across age bins, and how does that gap look at different ages?

4. Methods / Approaches

All resampling procedures use a fixed random seed (`np.random.seed(604)`) for reproducibility. Multi-ply subgroups were excluded from the analysis because none met the minimum-count threshold of 30 rows on both sides of the tested/untested split.

4.1 RQ1 Methods

H0: Untested entries do not have higher mean DOTS/TotalKg than tested entries within the same subgroup. Ha: Untested entries have higher mean DOTS/TotalKg (one-sided). For each qualifying subgroup `g`, let `mu_NT,g` and `mu_T,g` be the population means for untested and tested representative rows. The hypotheses apply to both TotalKg and DOTS: `H0: mu_NT,g = mu_T,g` (no difference between groups). `Ha: mu_NT,g > mu_T,g` (untested mean is higher; one-sided).

Using one representative row per `Name x Sex x Equipment x WeightClassKg x is_tested`, where repeated same-status entries are collapsed to mean `TotalKg` and mean `Dots` values first, subgroups with fewer than 30 representative rows on either side are excluded to keep group means stable. For each qualifying subgroup the untested-minus-tested gap is assessed three ways: **a 95% bootstrap CI** (10,000 resamples, percentile method, chosen to keep Monte Carlo noise small enough that the bootstrap CIs and permutation p-values are stable across runs); **a one-sided permutation test** (10,000 shuffles); and **a one-sided Welch t-test** as a parametric cross-check. Running 78 tests at `alpha` = 0.05 would produce several false positives by chance alone, so we apply a Bonferroni correction. Each permutation-test and Welch-test p-value is compared against `alpha / m = 0.05 / 78 = 0.000641` rather than 0.05, which controls the family-wise error rate across the full set of RQ1 subgroup tests (NIST/SEMATECH, n.d.).

4.2 RQ2 Methods

H0: DOTS-score bin and tested status are independent. Ha: DOTS-score bin and tested status are associated within each sex.

Using one representative row per `Name x Sex x is_tested`, where repeated same-status entries are collapsed to a mean `Dots` value first, the representative `Dots` values are split into five equal-width bins within sex (equal-width chosen for interpretability on the original DOTS scale) and a contingency table crosses bin by tested status. **A chi-squared test of independence** evaluates whether the tested-entry share differs across bins.

4.3 RQ3 Methods

Using one representative row per `Name x Sex x age_bin x is_tested` among rows with known age, the analysis assigns six age bins (`18–23`, `24–30`, `31–40`, `41–50`, `51–60`, `61+`) and collapses repeated same-status entries to a mean `TotalKg` value within each bin. The bins separate younger, prime-age, masters-age, and older lifters while keeping enough rows in each bin. It then summarizes mean `TotalKg` across those six age bins and uses bootstrap resampling (10,000 resamples) to estimate the untested-minus-tested gap in each `Sex x age_bin` pair.

5. Results and Analysis

5.1 RQ1 - Main Performance Comparison

Among the 78 subgroup pairs that met the minimum-count rule, the overall pattern was mixed because equipment types pulled in opposite directions (Table 1).

Table 1. RQ1 bootstrap 95% CI results by equipment type (78 qualifying subgroup pairs)

Equipment	Untested higher	Tested higher	No clear difference	Subgroups
Raw	14 (37%)	13 (34%)	11 (29%)	38
Wraps	19 (95%)	0 (0%)	1 (5%)	20
Single-ply	1 (5%)	19 (95%)	0 (0%)	20
All	**34 (44%)**	**32 (41%)**	**12 (15%)**	**78**

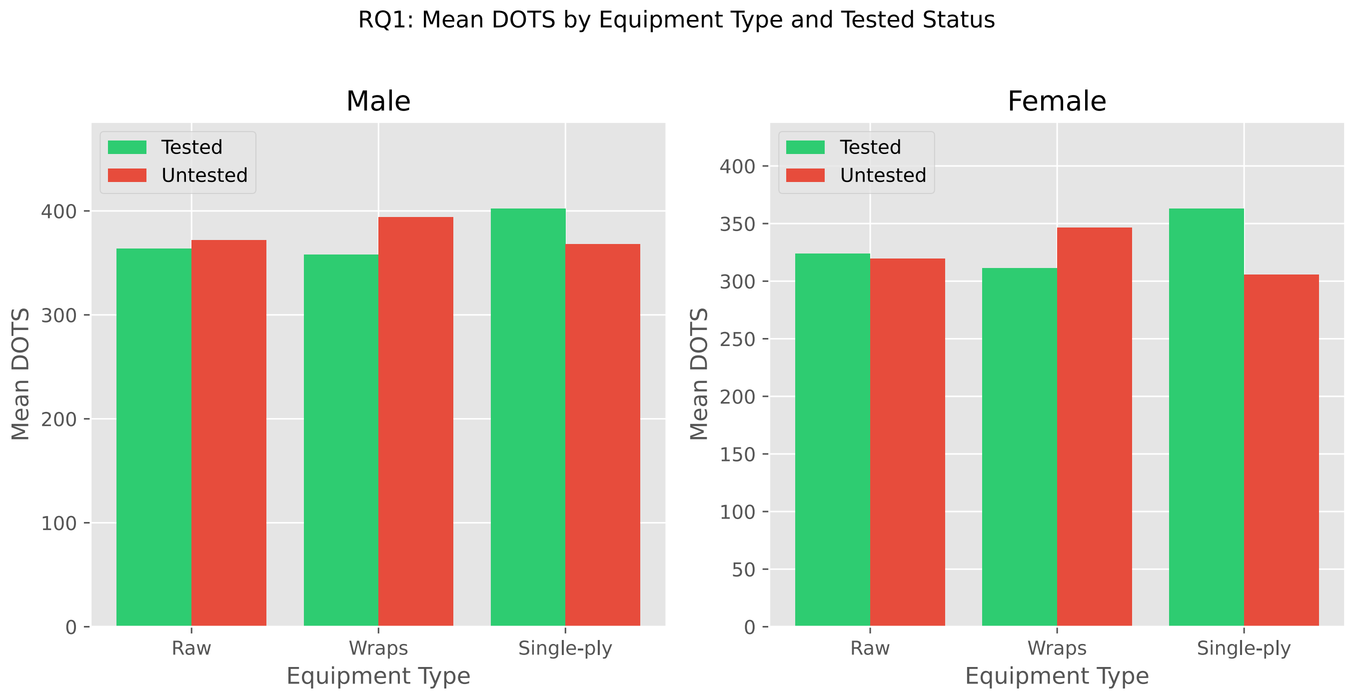


Figure 1. Overall mean DOTS by equipment type and tested status, shown separately for men (left) and women (right). Green bars represent tested entries; red bars represent untested entries.

Figure 1 shows the same split in plain form: the untested bar is visibly taller in Wraps, shorter in Single-ply, and nearly even in Raw. Multi-ply is absent because none of its subgroups met the minimum-count rule. After Bonferroni correction, 29 of the 78 pairs remained significant by both the permutation test and Welch t-test, for both TotalKg and DOTS. That is a stricter result than the count of 34 subgroups whose bootstrap intervals sat entirely above zero. Among the corrected-significant untested-higher subgroups, the TotalKg gap ranged from about 8 to 85 kg (median 38 kg).

5.2 RQ2 - Tested Status by DOTS-Score Bin

Table 2. RQ2 chi-squared results by sex

Sex	Chi-squared statistic	df	p-value	Expected-count check	Interpretation
Men	166.99	4	< 0.001	All >= 5 (min 83.46)	Reject H0
Women	510.37	4	< 0.001	All >= 5 (min 7.93)	Reject H0

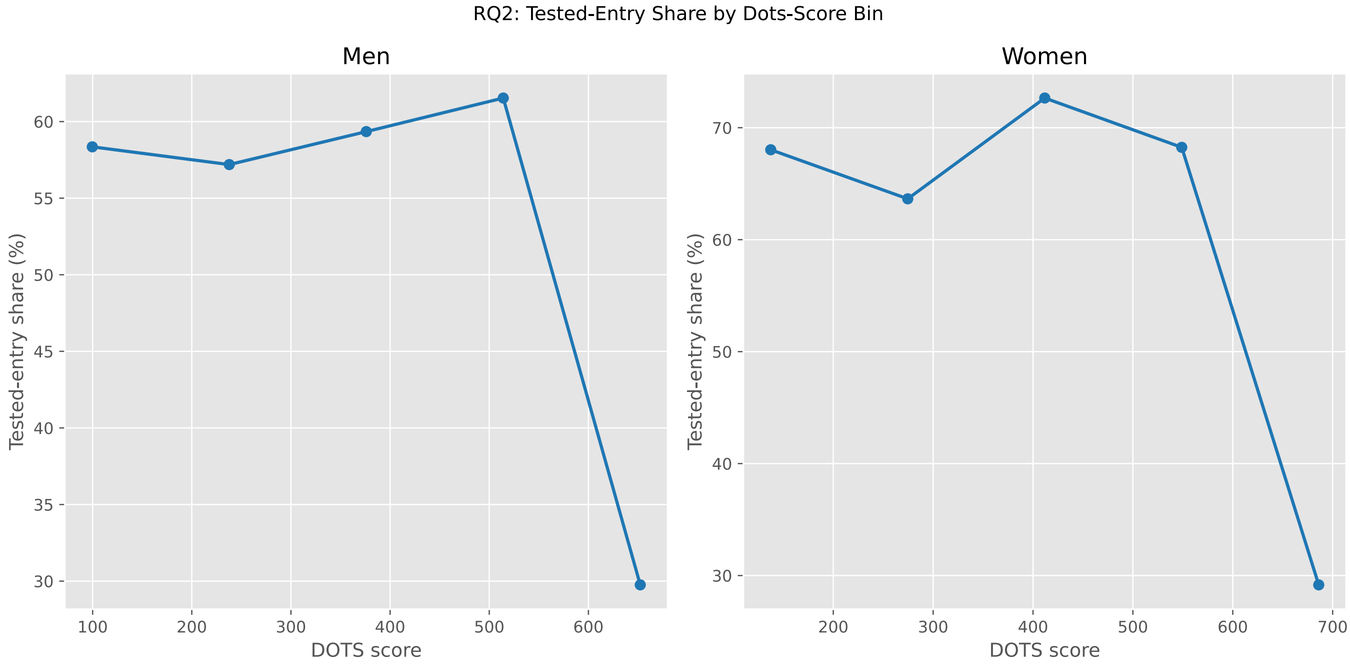
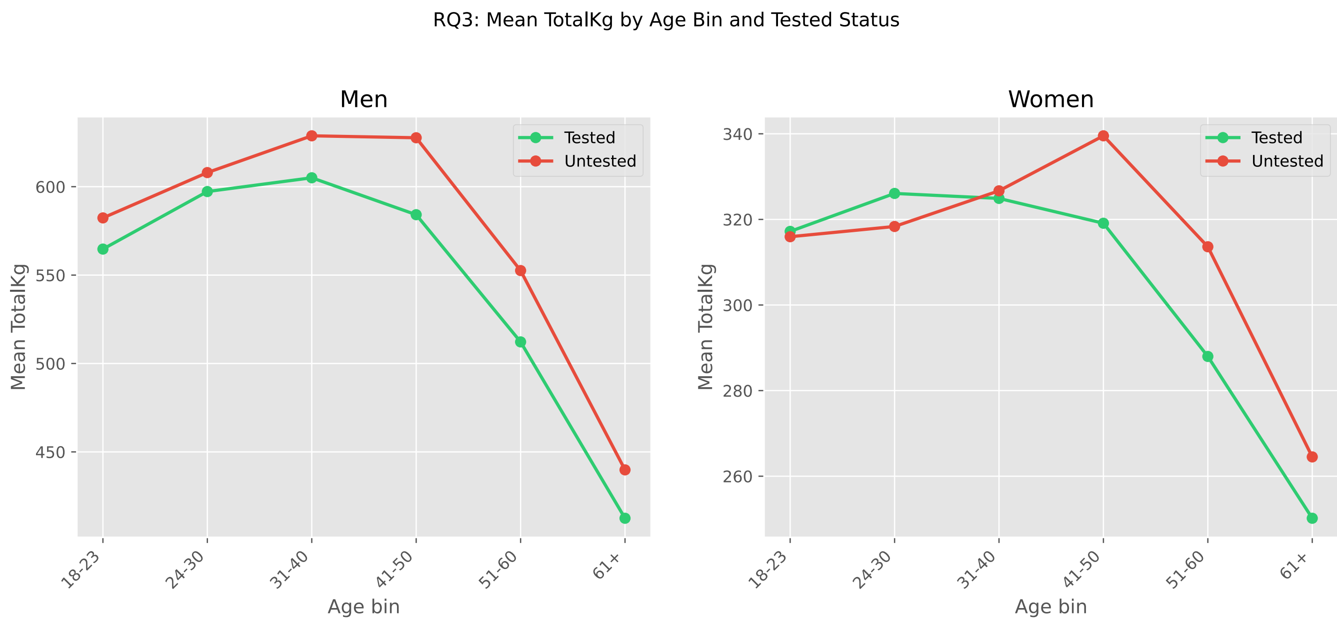


Figure 2. Tested-entry share (%) by DOTS-score bin, shown separately for men (left) and women (right).

For both sexes, the tested-entry share was fairly stable across the first four bins but dropped sharply to about 30% in the highest-DOTS bin, the only place where untested entries outnumbered tested ones.

5.3 RQ3 - Age Pattern Comparison



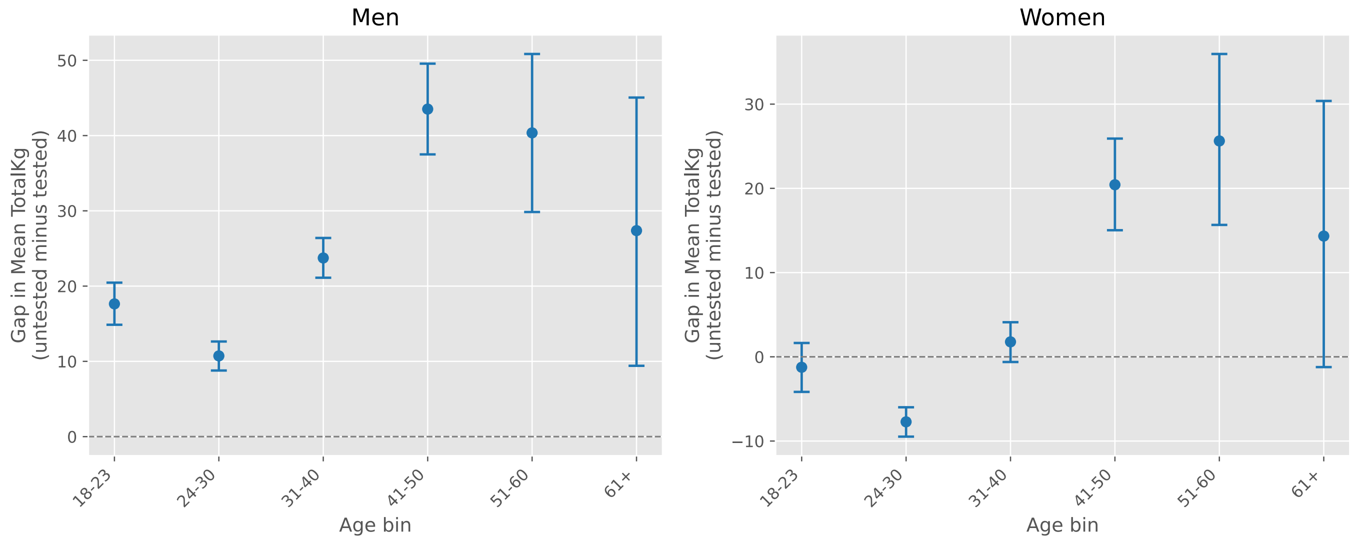


Figure 3. The upper row shows mean TotalKg by age bin and tested status, shown separately for men and women. In the lower row, each dot marks the observed untested-minus-tested gap, and the vertical bar spans the bootstrap 5th-to-95th percentile range for that bin (not a symmetric plus/minus error bar). The dashed line marks zero.

A bin counts as untested-higher when its bootstrap interval clears zero. For men, every age bin favored untested entries, with the gap smallest at 24-30 (~11 kg) and widest at 41-50 (~44 kg). For women, 24-30 was the only bin where tested entries clearly came out higher (~8 kg); 18-23, 31-40, and 61+ were inconclusive; the clearest untested advantage appeared at 41-50 and 51-60 (~20-26 kg).

6. Discussion / Insights

6.1 Significance

No single narrative fits the full picture. First, the Single-ply reversal seen in RQ1 probably comes down to which organizations run the biggest Single-ply divisions. Tested federations like the IPF have been running equipped lifting for decades, so their Single-ply fields tend to be large and deep with talent. Wraps are more popular in untested federations, where the biggest and most competitive fields form on that side instead. The gap flips direction depending on which side of the split has the stronger competition pool, which suggests that the structure of the sport itself, who competes where, may matter as much as whether drug testing is involved. In RQ2, the tested-entry share holding steady across most of the performance range and only dropping in the top bin suggests that tested and untested lifters overlap far more than the "natty or not" debate implies, with meaningful separation confined to the extreme high end. Finally, the widening RQ3 gap at older ages may not reflect a real shift in performance. Untested lifters still competing in their 40s and 50s tend to be the strongest of their cohort, while tested meets may retain a broader mix of older lifters, so the comparison stops being apples-to-apples.

6.2 Limitations

As noted in Section 2.2, the "Tested" field is a competition-context label, not proof of drug use or non-use. Athletes choose which context to compete in for many reasons, so the data cannot separate one explanation from another.

The cleaning described in Section 2.3 retained about 13% of rows and removed all crossover lifters, so the findings describe a specific slice of the sport, not all powerlifters; if the excluded entries follow different patterns, the conclusions may not generalize. Because crossover detection used name alone, some non-crossover lifters who share a name may have been incorrectly removed.

Finally, older meets tend to have more missing fields, and federation conventions around weight classes and equipment categories were not normalized. Some of the observed patterns may partly reflect which federations contributed the most data rather than a true tested/untested difference. RQ2's top DOTS bin is also sparse, with only 24 women (17 untested, 7 tested), so the ~30% tested share there is a directional clue rather than a precise estimate.

6.3 Improvements / Extensions

Three extensions could build on these findings: track the tested/untested gap over time, break the analysis down by federation, and study the crossover lifters excluded here so each athlete can serve as their own control.

7. Use of AI Tools

I used Claude (Anthropic) mainly as a learning tool and editing assistant. It helped me learn Python, troubleshoot notebook errors, and proofread. For example, it helped me understand tools like `pd.cut()` for age bins and the `alternative` argument in `scipy.stats.ttest_ind()` for one-sided tests. I chose the project topic, designed the research questions, and made all analysis decisions myself. Whenever Claude suggested anything outside the scope of the ICS 604 course, I either rejected it or adapted it. The notebook and written report were AI-assisted, but ultimately written by me.

8. Conclusion

This project asks whether untested powerlifting competition entries tend to show higher performance than tested entries.

RQ1 found that even after Bonferroni correction, the untested advantage depended heavily on equipment type, appearing consistently in Wraps but reversing in Single-ply. RQ2 found that untested entries only outnumbered tested entries in the highest-scoring DOTS bin, with little separation elsewhere. RQ3 found that the gap was widest in middle and older age bins and much clearer for men than for women.

The caveats in Section 6.2 apply throughout. The dramatically opposite patterns across equipment types suggest that competitive context may shape these results as much as testing policy does, and the results should not be read as a simple story about drug use.

References

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